VF Asia Footwear Laboratory Testing Manual

Test Method for Ross Flex	**Test method #: VF-D02** **REF #: TM-R- 030** **Page 1 of 4**
Date Issued: June,2019 **Revision #: 5** **Reason for revision: 1. Hydrolysis test temperature changed from 70°C to 60°C**	**Created by: Nila Babia Erazo**

PURPOSE AND SCOPE: To measure the ability of a material to resist cracking, as a result of being flexed repeatedly to a 90° angle.

REFERENCE: ASTM D1052

EQUIPMENT AND MATERIAL:

1. Ross Flex Machine A, meeting the requirements of ASTM D1052.
2. Piercing tool B with cutting edge of 2.5±0.02 mm in length, (SATRA STD 405 chisel holding jig with ASTM D 1052 chisel).
3. Digital calipers, capable of measuring to 0.01mm.
4. Freezing chamber capable of maintaining temperature up to (***negative***) - 25 ± 1°C or at the specified freezing temperature required by the brand.
5. Air-circulating oven, capable of maintaining 70 ± 1°C
6. Hydrolysis oven, capable of maintaining temperature of 60 ± 1°C and relative humidity at 95 ± 1%.
7. Rubber, thermoplastic rubber, plastic, insoles, painted midsoles, painted flexible polymers .

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory environment at 23.0 ± 2. 0 ˚C temperature and 55 ±10 % Relative Humidity.
2. Die-cut test samples 2.54cm in width by 15.2cm in length. Samples should have the following thickness:

Material	***Thickness (mm)***
Solid Natural Rubber, Synthetic Rubber, Resin Rubber	3.0 ± 0.2 mm
Thermoplastic Polyurethane, Polyester, Nylon, and similar material	3.0 ± 0.2 mm
Microcellular Rubber, EVA, and polyurethane	5.0 ± 0.2 mm
Thermoplastic rubber and similar material	7.0 ± 0.2 mm
Heel counter	as supplied

Test Method for Ross Flex	Test method #: VF-D02 REF #: TM-R- 030 Page 2 of 4
Date Issued: June,2019 Revision #: 5 Reason for revision: 1. Hydrolysis test temperature changed from 70°C to 60°C	Created by: Nila Babia Erazo

Notes:

- *For test specimens prepared from a manufactured article with a surface tread / pattern, or for cellular soling with a mold skin, do not remove the surface tread / pattern unless there's special requirement for the test. In case there's surface preparation, it should be noted and reported in the final results.*
- *If the material does not meet these thickness requirements, it shall be tested at the maximum possible thickness.*
- *Materials that are thicker than the specified thickness should be split or buffed to reduce the thickness so that it falls within the specified range.*

3. Position the test sample in the piercing tool such that the chisel tip is located over the center of the test sample and 62mm from one of the ends.
4. Push the chisel tip just far enough into the test sample so that a 2.5mm wide slit is produced throughout the entire thickness of the sample.

   Note: For test specimen with a surface pattern, ensure that the slit should be on the base NOT on the top of the tread / pattern.

5. So that the crack can be seen more easily, for colored materials, mark with silver pen the 2.5 mm slit while for light colored materials, mark with regular color pen.

Notes:

- No slit should be made in cellulose insole boards or painted midsoles**.**
- Do not push the chisel into the sample beyond its "flattened" end. If the rounded portion of the chisel does enter the sample, a slit with the incorrect geometry /shape will be produced and such samples must be discarded.

REPETITIONS: Test two (2) test samples per material.

TEST PROCEDURES:

A. Test at Room Temperature @ 23±2 ºC

1. Measure and record the thickness of the sample at three points across the width of the sample where the slit was made. ***Refer to VF Test Method VF- A03 for thickness measurement***.

Test Method for Ross Flex	Test method #: VF-D02 REF #: TM-R- 030 Page 3 of 4
Date Issued: June,2019 Revision #: 5 Reason for revision: 1. Hydrolysis test temperature changed from 70°C to 60°C	Created by: Nila Babia Erazo

2. Raise the holder arm of the Ross Flex machine to the horizontal position.
3. Insert the test sample through the clamp on the holder arm and the top rollers at the rear of the machine. The end of the test sample having the shorter distance to the slit should be positioned in the holder arm clamp.
4. Clamp the sample in the holder arm such that the slit is located directly over the area of the flexing rod and the sample is centered in the roller.
5. Lower the top rollers until they just touch the test sample, and lock in this position using the wing nut. The test sample should move freely under the top rollers.
6. Zero the counter and start the machine.
7. Examine the sample at regular intervals.
    - If the sample is to be run for a maximum of 100,000 cycles, check at 10,000, 25,000, 50,000, and 100,000 cycles.
    - In case a shorter test is to be run, for example at 25,000 cycles, the sample should be checked at every 5,000 cycles.

Note: ***When the machine will be stopped to examine the sample at the required test intervals, ensure that the holder arm should be positioned in the horizontal position so that there is no stress on the sample.***

8. Failure is defined as “when the slit has grown to a length of 5.0mm”. This evaluation should be made with the folder arm at an angle of approximately 45 degrees. If the crack growth is not in a straight line, measure the distance of the longest continuous crack, regardless of its direction.
    - For “thermo” heel counters, it is the plastic core which must crack to a length of 5.0mm.
    - For cellulose insole boards, failure occurs when the surface ply has cracked for a distance of 3.0mm.
    - For painted midsoles, failure occurs as soon as any cracks in the paint are observed.

B. Test at freezing temperature: @- 20^{0}C / - 25^{0}C / - 15 ^{0}C OR at specified freezing temperature

1. Follow steps in section A , **Test at Room Temperature**. The difference is that the test is carried out in the freezing chamber.

Note: ***Ensure that the chamber is close and allow to reach @ - 20^{0}C temperature or at the specified freezing temperature, before start testing.***

Test Method for Ross Flex	Test method #: VF-D02 REF #: TM-R- 030 Page 4 of 4
Date Issued: June,2019 Revision #: 5 Reason for revision: 1. Hydrolysis test temperature changed from 70°C to 60°C	Created by: Nila Babia Erazo

C. Test for Oven aging at 70 ± 1ºC for 3 or 7 days:

1. If aging is required prior to Ross Flex test, place the sample inside the heat aging oven at 70 ± 1ºC temperature for 3 or 7 days
2. After removal from the aging environment, allow the test sample to condition at 23±2 ºC and 55±10% relative humidity for at least 24 hours prior to Rose Flex testing.
3. Follow steps in section A , **Test at Room Temperature**.

D. Test for Hydrolysis aging at 60 ± 1ºC / 95% Relative humidity for 7 days:

1. If hydrolysis aging is required prior to Ross Flex test, place the sample inside the hydrolysis oven at 60 ± 1ºC temperature , 95% Relative humidity for 7 days.
2. After removal from the hydrolysis aging environment, allow the test sample to condition at 23±2 ºC and 55±10% relative humidity for at least 24 hours prior to Ross Flex testing.
3. Follow steps in section A , **Test at Room Temperature**.

CALCULATION:

Cut growth (%): (Lf - Li) / Li *100
Where: ***Li*** *= 2.5 mm* ***Lf*** *= Final cut growth*

REPORT: Percent (%)

Note: For cellulose, non-woven, or thermal plastic insole boards reporting is by a rating scale:

RATING SYSTEM FOR CELLULOSE, NON-WOVEN, OR THERMAL PLASTIC INSOLE BOARDS*

Rating	*Description*
1	Slight Creasing
2	Small cracks confined to top surface
3	Series of small cracks confined to top surface, limited impact on flexibility
4	Cracking into but <50% through, limited impact on flex integrity
5	Cracking into but <75%, flex integrity impacted (little or no flex resistance or permanently deformed)
6	>75% cracking or tearing of substrate.